

Appendix A: Classical Trigonometric and Hyperbolic Formulas

Circular and Hyperbolic Trigonometric Formulas

Basic Definitions

Circular Trigonometry:

$$\begin{aligned}\cos(x) &= \frac{e^{ix} + e^{-ix}}{2}, \quad \sin(x) = \frac{e^{ix} - e^{-ix}}{2i}, \\ \tan(x) &= \frac{\sin(x)}{\cos(x)} \quad \text{with } x \neq \frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z}, \\ \cot(x) &= \frac{\cos(x)}{\sin(x)} \quad \text{with } x \neq k\pi, \quad k \in \mathbb{Z}, \\ \tan(x) &= \frac{1}{i} \cdot \frac{e^{ix} - e^{-ix}}{e^{ix} + e^{-ix}}, \quad \cot(x) = -i \cdot \frac{e^{ix} + e^{-ix}}{e^{ix} - e^{-ix}}.\end{aligned}$$

Hyperbolic Trigonometry:

$$\begin{aligned}\cosh(x) &= \frac{e^x + e^{-x}}{2}, \quad \sinh(x) = \frac{e^x - e^{-x}}{2}, \\ \tanh(x) &= \frac{\sinh(x)}{\cosh(x)} = \frac{e^x - e^{-x}}{e^x + e^{-x}}, \\ \coth(x) &= \frac{\cosh(x)}{\sinh(x)} = \frac{e^x + e^{-x}}{e^x - e^{-x}} \quad \text{with } x \neq 0.\end{aligned}$$

Fundamental Identities

Circular:

$$\begin{aligned}\cos^2(x) + \sin^2(x) &= 1, \\ e^{ix} &= \cos(x) + i \sin(x), \\ \frac{d^2}{dx^2} e^{ix} &= -e^{ix}.\end{aligned}$$

Hyperbolic:

$$\begin{aligned}\cosh^2(x) - \sinh^2(x) &= 1, \\ e^x &= \cosh(x) + \sinh(x), \\ \frac{d^2}{dx^2} e^x &= e^x.\end{aligned}$$

Addition Formulas

Circular:

$$\begin{aligned}\cos(x + y) &= \cos x \cos y - \sin x \sin y, \\ \cos(x - y) &= \cos x \cos y + \sin x \sin y, \\ \sin(x + y) &= \sin x \cos y + \cos x \sin y, \\ \sin(x - y) &= \sin x \cos y - \cos x \sin y.\end{aligned}$$

Hyperbolic:

$$\begin{aligned}\cosh(x + y) &= \cosh x \cosh y + \sinh x \sinh y, \\ \cosh(x - y) &= \cosh x \cosh y - \sinh x \sinh y, \\ \sinh(x + y) &= \sinh x \cosh y + \cosh x \sinh y, \\ \sinh(x - y) &= \sinh x \cosh y - \cosh x \sinh y.\end{aligned}$$

Tangent Addition

Circular:

$$\begin{aligned}\tan(x+y) &= \frac{\tan x + \tan y}{1 - \tan x \tan y}, \\ \tan(x-y) &= \frac{\tan x - \tan y}{1 + \tan x \tan y}.\end{aligned}$$

Hyperbolic:

$$\begin{aligned}\tanh(x+y) &= \frac{\tanh x + \tanh y}{1 + \tanh x \tanh y}, \\ \tanh(x-y) &= \frac{\tanh x - \tanh y}{1 - \tanh x \tanh y}.\end{aligned}$$

Product-to-Sum Formulas**Circular:**

$$\begin{aligned}\cos x \cos y &= \frac{1}{2}[\cos(x-y) + \cos(x+y)], \\ \sin x \sin y &= \frac{1}{2}[\cos(x-y) - \cos(x+y)], \\ \sin x \cos y &= \frac{1}{2}[\sin(x-y) + \sin(x+y)].\end{aligned}$$

Hyperbolic:

$$\begin{aligned}\cosh x \cosh y &= \frac{1}{2}[\cosh(x+y) + \cosh(x-y)], \\ \sinh x \sinh y &= \frac{1}{2}[\cosh(x+y) - \cosh(x-y)], \\ \sinh x \cosh y &= \frac{1}{2}[\sinh(x+y) + \sinh(x-y)].\end{aligned}$$

Addition and Subtraction Formulas for Trigonometric and Hyperbolic Functions**Trigonometric Identities**

$$\begin{aligned}\cos(x+y) &= \cos x \cos y - \sin x \sin y, \\ \cos(x-y) &= \cos x \cos y + \sin x \sin y, \\ \sin(x+y) &= \sin x \cos y + \cos x \sin y, \\ \sin(x-y) &= \sin x \cos y - \cos x \sin y, \\ \tan(x+y) &= \frac{\tan x + \tan y}{1 - \tan x \tan y} = \frac{\sin(x+y)}{\cos x \cos y}, \\ \tan(x-y) &= \frac{\tan x - \tan y}{1 + \tan x \tan y} = \frac{\sin(x-y)}{\cos x \cos y}.\end{aligned}$$

where $x, y \notin \left\{\frac{\pi}{2} + k\pi : k \in \mathbb{Z}\right\}$ for the tangent identities.

Hyperbolic Identities

$$\begin{aligned}\cosh(x+y) &= \cosh x \cosh y + \sinh x \sinh y, \\ \cosh(x-y) &= \cosh x \cosh y - \sinh x \sinh y, \\ \sinh(x+y) &= \sinh x \cosh y + \cosh x \sinh y, \\ \sinh(x-y) &= \sinh x \cosh y - \cosh x \sinh y, \\ \tanh(x+y) &= \frac{\tanh x + \tanh y}{1 + \tanh x \tanh y} = \frac{\sinh(x+y)}{\cosh x \cosh y}, \\ \tanh(x-y) &= \frac{\tanh x - \tanh y}{1 - \tanh x \tanh y} = \frac{\sinh(x-y)}{\cosh x \cosh y}.\end{aligned}$$

Valid where all terms are defined.

Double Angle and Power-Reduction Formulas**Trigonometric Identities**

$$\begin{aligned}\sin(2x) &= 2 \sin x \cos x, \\ \cos(2x) &= \cos^2 x - \sin^2 x \\ &= 2 \cos^2 x - 1 \\ &= 1 - 2 \sin^2 x, \\ \cos^2 x &= \frac{1 + \cos(2x)}{2}, \\ \sin^2 x &= \frac{1 - \cos(2x)}{2}.\end{aligned}$$

Hyperbolic Identities

$$\begin{aligned}\sinh(2x) &= 2 \sinh x \cosh x, \\ \cosh(2x) &= \cosh^2 x + \sinh^2 x \\ &= 2 \cosh^2 x - 1 \\ &= 2 \sinh^2 x + 1, \\ \cosh^2 x &= \frac{\cosh(2x) + 1}{2}, \\ \sinh^2 x &= \frac{\cosh(2x) - 1}{2}.\end{aligned}$$

Formulas in Terms of $t = \tan\left(\frac{x}{2}\right)$

Trigonometric Identities

Let $t = \tan\left(\frac{x}{2}\right)$, $x \neq (2k+1)\pi$, $k \in \mathbb{Z}$.

$$\cos x = \frac{1-t^2}{1+t^2},$$

$$\sin x = \frac{2t}{1+t^2},$$

$$\tan x = \frac{2t}{1-t^2}, \quad x \neq \frac{\pi}{2} + k\pi,$$

$$e^{ix} = \frac{1+it}{1-it},$$

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i} = \frac{2t}{1+t^2},$$

$$\cos x = \frac{e^{ix} + e^{-ix}}{2} = \frac{1-t^2}{1+t^2}.$$

Hyperbolic Identities

Let $t = \tanh\left(\frac{x}{2}\right)$. Then:

$$\cosh x = \frac{1+t^2}{1-t^2},$$

$$\sinh x = \frac{2t}{1-t^2},$$

$$\tanh x = \frac{2t}{1+t^2},$$

$$e^x = \frac{1+t}{1-t},$$

$$\sinh x = \frac{e^x - e^{-x}}{2} = \frac{2t}{1-t^2},$$

$$\cosh x = \frac{e^x + e^{-x}}{2} = \frac{1+t^2}{1-t^2}.$$

Triple Angle Formulas (Circular vs. Hyperbolic)

Circular functions:

- $\cos(3x) = 4\cos^3 x - 3\cos x$
- $\sin(3x) = 3\sin x - 4\sin^3 x$
- $\tan(3x) = \frac{3\tan x - \tan^3 x}{1 - 3\tan^2 x}$

Valid for $x \notin \left\{\frac{\pi}{6} + \frac{k\pi}{3} : k \in \mathbb{Z}\right\}$

Hyperbolic functions:

- $\cosh(3x) = 4\cosh^3 x - 3\cosh x$
- $\sinh(3x) = 3\sinh x + 4\sinh^3 x$
- $\tanh(3x) = \frac{3\tanh x + \tanh^3 x}{1 + 3\tanh^2 x}$

Defined for all $x \in \mathbb{R}$

Summation Identities of the Form $\sum_{k=0}^n f(x + ky)$

Trigonometric Case

$$\sum_{k=0}^n \cos(x + ky) = \frac{\sin\left(\frac{(n+1)y}{2}\right) \cdot \cos\left(x + \frac{ny}{2}\right)}{\sin\left(\frac{y}{2}\right)}$$

$$\sum_{k=0}^n \sin(x + ky) = \frac{\sin\left(\frac{(n+1)y}{2}\right) \cdot \sin\left(x + \frac{ny}{2}\right)}{\sin\left(\frac{y}{2}\right)}$$

Valid for $y \notin \pi\mathbb{Z}$ in the trigonometric case, and $y \neq 0$ in the hyperbolic case.

Hyperbolic Case

$$\sum_{k=0}^n \cosh(x + ky) = \frac{\sinh\left(\frac{(n+1)y}{2}\right) \cdot \cosh\left(x + \frac{ny}{2}\right)}{\sinh\left(\frac{y}{2}\right)}$$

$$\sum_{k=0}^n \sinh(x + ky) = \frac{\sinh\left(\frac{(n+1)y}{2}\right) \cdot \sinh\left(x + \frac{ny}{2}\right)}{\sinh\left(\frac{y}{2}\right)}$$

Generalized de Moivre Formulas for Rational Powers

These identities are valid for all $m \in \mathbb{Z}$.

Circular functions:

$$(\cos x + i \sin x)^m = \cos(mx) + i \sin(mx)$$

$$(\cos x - i \sin x)^m = \cos(mx) - i \sin(mx)$$

Hyperbolic functions:

$$(\cosh x + \sinh x)^m = \cosh(mx) + \sinh(mx)$$

$$(\cosh x - \sinh x)^m = \cosh(mx) - \sinh(mx)$$

Derivatives of Circular and Hyperbolic Functions**Circular Trigonometric Functions**

$$\begin{aligned} \frac{d}{dx} \cos x &= -\sin x, & \text{for all } x \in \mathbb{R}, \\ \frac{d}{dx} \sin x &= \cos x, & \text{for all } x \in \mathbb{R}, \\ \frac{d}{dx} \tan x &= \frac{1}{\cos^2 x}, & x \in \mathbb{R} \setminus \left\{ \frac{\pi}{2} + k\pi \right\}, \\ \frac{d}{dx} \cot x &= -\frac{1}{\sin^2 x}, & x \in \mathbb{R} \setminus \{k\pi\}. \end{aligned}$$

Inverse Circular Functions

$$\begin{aligned} \frac{d}{dx} \arcsin x &= \frac{1}{\sqrt{1-x^2}}, & x \in]-1, 1[, \\ \frac{d}{dx} \arccos x &= -\frac{1}{\sqrt{1-x^2}}, & x \in]-1, 1[, \\ \frac{d}{dx} \arctan x &= \frac{1}{1+x^2}, & x \in \mathbb{R}, \\ \frac{d}{dx} \cot^{-1} x &= -\frac{1}{1+x^2}, & x \in \mathbb{R}. \end{aligned}$$

Hyperbolic Trigonometric Functions

$$\begin{aligned} \frac{d}{dx} \cosh x &= \sinh x, & \text{for all } x \in \mathbb{R}, \\ \frac{d}{dx} \sinh x &= \cosh x, & \text{for all } x \in \mathbb{R}, \\ \frac{d}{dx} \tanh x &= \frac{1}{\cosh^2 x}, & \text{for all } x \in \mathbb{R}, \\ \frac{d}{dx} \coth x &= -\frac{1}{\sinh^2 x}, & \text{for all } x \in \mathbb{R} \setminus \{0\}. \end{aligned}$$

Inverse Hyperbolic Functions

$$\begin{aligned}
 \frac{d}{dx} \operatorname{argcosh} x &= \frac{1}{\sqrt{x^2 - 1}}, & x \in]1, +\infty[, \\
 \frac{d}{dx} \operatorname{argsinh} x &= \frac{1}{\sqrt{x^2 + 1}}, & x \in \mathbb{R}, \\
 \frac{d}{dx} \operatorname{argtanh} x &= \frac{1}{1 - x^2}, & x \in]-1, 1[, \\
 \frac{d}{dx} \operatorname{argcoth} x &= \frac{1}{1 - x^2}, & x \in]-\infty, -1[\cup]1, +\infty[.
 \end{aligned}$$

Logarithmic Forms of Inverse Hyperbolic Functions

$$\begin{aligned}
 \operatorname{argcosh} x &= \ln(x + \sqrt{x^2 - 1}), & x \in]1, +\infty[, \\
 \operatorname{argsinh} x &= \ln(x + \sqrt{x^2 + 1}), & x \in \mathbb{R}, \\
 \operatorname{argtanh} x &= \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right), & x \in]-1, 1[, \\
 \operatorname{argcoth} x &= \frac{1}{2} \ln \left(\frac{x+1}{x-1} \right), & x \in]-\infty, -1[\cup]1, +\infty[.
 \end{aligned}$$